

Theme Y Demo Frame - Beam B-RB (interior, span 6.0 m)

Code: ACI 318-19 | Designer: femsolver | 2026-05-30

ALL CHECKS PASS

Flexure check (positive moment)

Mid-span positive moment check for the interior roof beam under factored gravity loading.

Inputs

Symbol	Value	Description
f_c'	3.0000e+07 Pa	Concrete compressive strength
f_y	4.2000e+08 Pa	Rebar yield strength
b	0.3 m	Beam width
d	0.45 m	Effective depth
rho	0.012	Tension steel ratio
w_u	3e+04 N/m	Factored line load
L	6 m	Span

Calculations

$A_s = \rho \cdot b \cdot d$

$A_s = 0.00162 \text{ m}^2$ (Tension steel area)

$a = A_s \cdot f_y / (0.85 \cdot f_c' \cdot b)$

$a = 0.08894 \text{ m}$ (Equivalent stress-block depth)

$\phi \cdot A_s \cdot f_y \cdot (d - a/2)$

$\phi M_n = 2.4833\text{e}+05 \text{ N.m}$ (Design moment capacity)

$w_u \cdot L^2 / 8$

$M_u = 1.3500\text{e}+05 \text{ N.m}$ (Factored midspan moment)

Checks

Check	Demand	Capacity	DCR	Status	Clause
Flexure (positive moment)	1.3500e+05 N.m	2.4833e+05 N.m	0.544	PASS	ACI 318-19 22.2.2